

the HDR T&R plans for the remaining fractions are adapted to be less restrictive for critical structure sparing. The process is repeated if for any of the T&R plans the dose for the OARs exceeds the maximum allowable dose. Finally, the calculated dose values from the EBRT and the 5 HDR T&R plans are introduced in the BED-EQD2 table and the doses to the critical structures are compared with the corresponding tolerance values. A MIM final composite is also performed to estimate the DVH distribution for the critical structures. If less than 5 T&R implants are performed, the treatment may continue with an SBRT boost. We extrapolated the methodology described above for such EBRT-HDR-SBRT cases. We follow the approach as detailed above to generate realistic HDR T&R plans. After the HDR approach is discontinued, we export to MIM the EBRT plan(s) and all delivered HDR T&R plans. A BED composite plan is done and a DVH analysis is performed to estimate the cumulative D2cc for the critical structures. The maximum allowable doses for the critical structures are then determined and used as constraints for the SBRT plan.

Results: We applied the methodology described above for a few cases. One representative case is described below. The patient was treated with 28 fractions of EBRT to 50.4 Gy, using an AP-PA approach. The D2cc for bladder, rectum, and sigmoid in the EBRT plan were 59.4, 59.3, and 59.3 Gy, respectively. By using these values in the BED-EQD2 table, the bladder/rectum/sigmoid tolerances for the HDR T&R approach become 4.1, 2.5, and 2.5 Gy/fraction, respectively. To do a more realistic estimation of those HDR tolerances we contoured in the EBRT plan new partial bladder, rectum, and sigmoid structures, including only the areas around the future HDR treatment region. The D2cc for these new structures were reduced to 55 Gy for the EBRT, resulting in larger HDR tolerances (4.6, 3.2, and 3.2 Gy/fraction for bladder, rectum, and sigmoid, respectively).

The final cumulative bladder/rectum/sigmoid D2cc BED for the EBRT and 5 HDR T&R, obtained from our table (worst case scenario) were 132, 117, and 117 when using the partial structures. If we considered the original bladder/rectum/sigmoid structures these values were 142, 127, and 127, with the rectum and sigmoid doses slightly higher than the stated tolerances. The final MIM composite DVH showed a BED of 120, 107, and 107 for bladder, rectum, and sigmoid, respectively.

Conclusions: The approach was successfully used for a few EBRT-HDR and EBRT-HDR-SBRT special cases.

PO33

Evaluate and Compare the Doses Differences between Line Source and Point Source Calculation Methods for Episcleral Plaque Brachytherapy

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Purpose: Episcleral plaque brachytherapy is a proven modality for treatment of medium-sized ocular melanoma tumors. Radiation dose calculation for this technique was previously performed using the Task Group 43 point-source system. However, the newer Task Group 129 is now recommending using the line-source system. The purpose of this study is to evaluate the dose differences between the two calculation methods on preservation of vision.

Materials and Methods: Two programs were created using Matlab for line source and point source approximation methods. The comparison was solely based on the doses differences between the two methods ignoring all other factors such as heterogeneous correction, backscatter and shielding effects from the bowl backing, the Silastics insert and the collimator clips. The doses were calculated for 106 patients treated with I-125 Amersham model 6711 and Best Medical model 2301 seeds. Differences between both methods were compared and correlated with preservation of visual acuity.

Results: The average percent doses differences were calculated for 57 patients treated with Amersham model 6711 showing 0.39% ($\pm 0.77\%$) at apex, 0.31% ($\pm 0.76\%$) at 5mm point, 1.05% ($\pm 0.63\%$) at sclera, 0.21%

($\pm 0.53\%$) at eye center, 0.104% ($\pm 0.32\%$) at contralateral retina, -0.87% ($\pm 1.17\%$) at macula, -0.03% ($\pm 0.38\%$) at lens and -1.08% ($\pm 0.8\%$) at optic disk. For the 49 patients treated with Best Medical model 2301, the average percent doses differences were 0.49% ($\pm 0.73\%$) at apex, 0.44% ($\pm 0.74\%$) at 5mm point, 1.31% ($\pm 0.63\%$) at sclera, 0.33% ($\pm 0.49\%$) at eye center, 0.33% ($\pm 0.30\%$) at contralateral retina, -2.8% ($\pm 1.97\%$) at macula, 2.03% ($\pm 0.34\%$) at lens, and 2.04% ($\pm 1.03\%$) at optic disk. Although, the maximum dose difference was found to be 7% at the macula, there were no significant differences between the doses calculated by point or source methods. Patients with retained visual acuity tended to received lower doses to the macula ($p=0.001$), and optic disk ($p<0.001$), regardless of dose calculation method.

Conclusions: The doses calculated by point and line source methods showed similar results for both tumor and normal structures. Furthermore, doses differences calculated by the two methods were not associated with any clinically significant benefit or detriment in visual preservation. Future research on doses calculation should take in account the heterogeneous correction, backscatter and shielding effects from the bowl backing, the Silastics inserts and the collimator lips.

PO34

Study of Dosimetric and Spatial Variations Due To Applicator Positioning During Inter-fraction High-dose-rate Brachytherapy in the Treatment of Carcinoma of Cervix - A 3D Dosimetric Analysis

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Purpose: This study was designed to study dosimetric and spatial variations in interfraction applicator positioning in high-dose-rate (HDR) brachytherapy.

Materials and Methods: This study includes 10 retrospective cervical cancer patients (Ca Cx). Each patient received 5 fractions of high-dose-rate (HDR) intracavitary brachytherapy. In all 50 plans, dose was originally prescribed to point A (ICRU-38), called Pt A. Post treatment, tumor volumes (HR-CTV) were drawn on CT images of each plan by the treating physician. Plan of the first implant is considered as a reference, and all subsequent CT image data sets were rigidly registered on first implant's CT data set with respect to the cervix or flange location. Another Point A, called Pt A ABS was defined on first plan and all subsequent plans, with reference to the ABS 2011 recommendations. Pt. A and Pt. A ABS of the first plan was carried over to subsequent plans keeping its original location. On subsequent plans, Pt. A and Pt. A ABS had its original location but dose distribution is from different subsequent plans (plan 2-5). Registration properties were recorded for all 50 plans that includes X, Y, Z DICOM offset, Rotational, Translational values and A D90 doses were also recorded.

Results: The mean angle of rotation on X, Y and Z axes are 0.49 ± 1.60 deg, -0.67 ± 2.31 deg and -0.10 ± 1.82 deg, respectively. While mean translational motion on X, Y and Z axes are -0.35 ± 3.22 cm, 0.53 ± 2.57 cm and 1.64 ± 9.26 cm, respectively. Mean normalized D90, mean HR-CTV and mean %IDL are 1.18 ± 0.20 , 23.25 ± 8.3 cc and 87.36 ± 20.26 cc, respectively. The mean of point A doses of each patient is compared with that of other patients using the method of the analysis of variance (ANOVA). The mean of all 5 plans of each patient do not have statistically significant difference ($p=0.225$). The difference between mean doses of point A ABS ($p=0.011$), mean doses point A registered ($p=0.005$) and mean doses of point A ABS registered ($p=0.0032$), were statistically different. The comparison between the doses of the point As, defined using ICRU-38, ABS 2011 and computed by registering on plan of first implant, are statistically different ($p < 0.05$). The mean HR-CTV of each patient were fitted with normalized D90 (ND90) and %IDL data using the method of least square fit. The ND90 has negative correlation with HR-CTV with correlation coefficient of 0.7253, while 100% IDL